CHAPTER 2

Ergodicity and Mixing (Basic ideas)

The concept of ergodicity is a very important one in dynamical systems, yet it turns out to be surprisingly difficult to establish if a system is or not ergodic and very few examples have been fully analyzed. Nonetheless, in this chapter we will see that a very simple idea introduced by Hopf [47, 48] allows to discuss the ergodicity in some special cases. The relevance of Hopf's idea is that, properly generalized, it allows to prove ergodicity in a vast class of systems. Much in the following chapters will deal with such a generalization.

2.1 A Basic Example

To explain the Hopf approach we will study a very simple case: a slight generalization of Arnold's cat, see Examples 1.1.1. Let $T : \mathbb{T}^2 \to \mathbb{T}^2$ (here by $\mathbb{T}^2$ we mean $\mathbb{R}^2 \mod 1$) be defined by

$$T \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 1 & a \\ a & 1+a^2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \mod 1 \tag{2.1.1}$$

It is obvious that if $a \in \mathbb{Z}$, then T is well defined and it is a linear automorphism of $\mathbb{T}^2$. Moreover, for all $x \in \mathbb{T}^2$

$$D_x T = \begin{pmatrix} 1 & a \\ a & 1+a^2 \end{pmatrix} \equiv L.$$

Since $\det L = 1$, Lebesgue measure is preserved. It is immediate to see that there exists $\lambda > 1$; v_+, $v_- \in \mathbb{R}^2$:

$$\begin{aligned} Lv_+ &= \lambda v_+ \\ Lv_- &= \lambda^{-1} v_-. \end{aligned}$$

We will call v_+ the unstable eigenvector (direction) and v_- the stable eigenvector (direction). Remark that, since $L^* = L$, $\langle v_+, v_-\rangle = 0$.

The dynamical system just described is a basic model of hyperbolic systems (see next chapter) and will appear in various disguises in this book.

Proposition 2.1.1 *The Arnold cat is ergodic.*

Sections 2.1.1 and 2.2.1 contain two different proofs of the above proposition.

2.1.1 An algebraic proof

A first idea to studying the ergodic properties of this system is to imitate what we have done for the Rotations (Examples 1.5.1) and the Dilations (Examples 1.7.1): use Fourier series. Let us see how such an approach would work.

Let $f,\ g \in C^{(m)}(\mathbb{T}^2)$, then[1]

$$f \circ T^n(x) = \sum_{k\in\mathbb{Z}^2} e^{2\pi i\langle k,\, L^n x\rangle} f_k = \sum_{k\in\mathbb{Z}^2} e^{2\pi i\langle k,\, x\rangle} f_{L^{-n}k},$$

so

$$\begin{aligned}\int_{\mathbb{T}^2} f\circ T^{2n} g = \int_{\mathbb{T}^2} f\circ T^n g\circ T^{-n} &= \sum_{k\in\mathbb{Z}^2} f_{L^{-n}k} g_{L^n k}\\ &= f_0 g_0 + \sum_{k\in\mathbb{Z}^2\setminus\{0\}} f_{L^{-n}k} g_{L^n k}.\end{aligned}$$

It is well known [75] that $f \in C^{(m)}(\mathbb{T}^2)$ implies[2]

$$|f_k| \le \frac{\|f^{(m)}\|_1}{\|k\|^m} \text{ for } k \ne 0$$

hence

$$\left|\sum_{k\in\mathbb{Z}^2\setminus\{0\}} f_{L^{-n}k} g_{L^n k}\right| \le \sum_{k\in\mathbb{Z}^2\setminus\{0\}} \frac{\|f^{(m)}\|_1 \|g^{(m)}\|_1}{\|L^{-n}k\|^m \|L^n k\|^m}.$$

For each $k \in \mathbb{Z}^2$ holds $\|k\|^2 = \langle k,\, v^+\rangle^2 + \langle k,\, v^-\rangle^2$ hence one of the two terms must be larger than $\|k\|^2/2$.[3] Moreover if $k \ne 0$ $\|L^n k\| \ge 1$ for each $n \in \mathbb{Z}$. Using the above facts it yields

[1] Note that $e^{2\pi i\langle k, T^n x\rangle} = e^{2\pi i\langle k, L^n x\rangle}$.

[2] Here for $\|f^{(m)}\|_1$ we mean $\sup_{\substack{i+j=m\\ i,j\ge 0}} \frac{1}{(2\pi)^m}\int_{\mathbb{T}^2} |\partial^i_{x_1}\partial^j_{x_2} f| dx_1 dx_2$; and $\|k\| = \sqrt{k_1^2 + k_2^2}$.

[3] Here we have normalized the eigenvalues so that $\|v^\pm\| = 1$.

$$\left|\sum_{k\in\mathbb{Z}^2\setminus\{0\}} f_{L^{-n}k}g_{L^nk}\right| \leq \sum_{k\in\mathbb{Z}^2\setminus\{0\}} \frac{\|f^{(m)}\|_1\|g^{(m)}\|_1 2^{m/2}}{\lambda^{nm}\|k\|^m}$$
$$\leq \text{const.}\|f^{(m)}\|_1\|g^{(m)}\|_1\lambda^{-nm},$$

where the constant does not depend on f or g and we have assumed $m \geq 3$ to insure the convergence of the series.

Accordingly, for each $f,\ g \in C^{(3)}(\mathbb{T}^2)$ we have

$$\left|\int_{\mathbb{T}^2} f\circ T^n g - \int_{\mathbb{T}^2} f \int_{\mathbb{T}^2} g\right| \leq \text{const.}\|f^{(3)}\|_1\|g^{(3)}\|_1\lambda^{-3n/2}.$$

To obtain the final result we need an approximation argument. If $f, g \in L^2(\mathbb{T}^2)$ we can choose $f_n, g_n \in C^{(3)}(\mathbb{T}^2)$ such that they converge to f and g, respectively, in L^2.

Then, for each $\varepsilon \geq 0$, choose $m \in \mathbb{N}$ such that

$$\|f - f_m\|_2 + \|g - g_m\|_2 \leq \varepsilon.$$

Accordingly,

$$\left|\int_{\mathbb{T}^2} f\circ T^n g - \int_{\mathbb{T}^2} f \int_{\mathbb{T}^2} g\right| \leq \left|\int_{\mathbb{T}^2} f_m\circ T^n g_m - \int_{\mathbb{T}^2} f_m \int_{\mathbb{T}^2} g_m\right|$$
$$+ 2\|f - f_m\|_2\|g\|_2 + 2\|f_m\|_2\|g - g_m\|_2$$
$$\leq 2(\|g\|_2 + \|f\|_2)\varepsilon + \varepsilon,$$

where we have chosen n large enough depending on m and ε. We have just proven mixing.

The above result is certainly rather satisfactory: non only it proves the mixing–hence the ergodicity–of the map but gives an explicit estimate on the rate of decay and shows how such a rate depends on the regularity of the functions.[4] Therefore, an eventual critique can not concern the type of result but only the method; indeed the method does have a shortcoming.

The use of Fourier series is strictly related to the group structure of $\mathbb{T}^2$ and the linearity of the map. Clearly in more general systems, where both such properties may fail, such a technique has no hope whatsoever of being applied.[5] In some sense, much of the theory of hyperbolic systems may be

[4] In fact, the obtained estimate it is not optimal: using the Diofantine properties of the stable and unstable directions a better estimate can be obtained (see Problem 2.1).

[5] In fact, there are very few cases in which this type of ideas has produced relevant results, noticeably the case of geodesic flows on surfaces of constant negative curvature [1].

view as an attempt to find an alternative proof of the above facts. Such a proof must be *dynamical* meaning that it must use properties of the dynamics and as little as possible of the structure of the space.

The best way to gain a real feeling of what is meant by *dynamical* is to see such type of arguments in action.

2.2 An Idea by Hopf

The following argument, due to Hopf [47], [48] is exactly such a dynamical proof of ergodicity. Let $f : \mathbb{T}^2 \to \mathbb{R}$ be a continuous function. We want to prove that for almost every $x \in \mathbb{T}^2$ the time averages converge as $n \to +\infty$ to the average value of f, i.e., $\int_{\mathbb{T}^2} f d\mu$. Once this is established one can obtain the same property for all integrable functions by an approximation argument, this proves ergodicity due to the characterization provided by Theorem 1.6.6 (see also Problem 1.27). From Birkhoff Ergodic Theorem (BET) we know that the time averages converge almost everywhere to a function $f^+ \in L^1(\mathbb{T}^2, \mu)$ which is invariant on the orbits of T, i.e., $f^+ \circ T = f^+$, and has the same average value as f, i.e., $\int f^+ d\mu = \int f d\mu$. Further, applying BET to f and T^{-1} we obtain that the time averages in the past

$$\frac{f(x) + f(T^{-1}x) + \cdots + f(T^{-n+1}x)}{n}$$

converge almost everywhere as $n \to +\infty$ to $f^- \in L^1(\mathbb{T}^2, \mu)$, $f^- \circ T = f^-$ and $\int f^- d\mu = \int f d\mu$.

The next Lemma is part of the usual magic of the ergodic theory.

Lemma 2.2.1 *The functions f^+ and f^- coincide almost everywhere.*

Proof. Let

$$\mathcal{A}_+ = \{x \in \mathbb{T}^2 \mid f_+(x) > f_-(x)\};$$

by definition $\mathcal{A}_+$ is an invariant set, hence

$$\int_{\mathcal{A}_+} [f_+(x) - f_-(x)]\, d\mu(x) = \int_{\mathcal{A}_+} f(x) d\mu(x) - \int_{\mathcal{A}_+} f(x) d\mu(x) = 0$$

which implies $\mu(\mathcal{A}_+) = 0$ and $f_+ \leq f_-$ μ-almost everywhere. The same argument, this time applied to the set $\mathcal{A}_- = \{x \in \mathbb{T}^2 \mid f_-(x) > f_+(x)\}$, implies the converse inequality. □

2.2.1 A dynamical proof

For $x \in \mathbb{T}^2$ let us denote by $W^u(x)$ ($W^s(x)$) the line in $\mathbb{T}^2$ passing through x and having the direction of the unstable eigenvector (the stable eigenvector), i.e., the eigenvector with eigenvalue λ (λ^{-1}). We call $W^u(x)$ ($W^s(x)$) the unstable (stable) leaf (or manifold) of x. The leaves of x have the following property. If $y \in W^u(x)$ ($y \in W^s(x)$) then the distance (computed along the leaf)

$$d(T^n y, T^n x) = \lambda^{-|n|} d(y, x) \to 0 \quad \text{as} \quad n \to -\infty(+\infty).$$

Hence for $y, z \in W^{u(s)}(x)$

$$|f(T^n y) - f(T^n z)| \to 0 \quad \text{as} \quad n \to -\infty(+\infty).$$

It follows that for $y, z \in W^{u)}(x)$ either $f^-(y)$ and $f^-(z)$ are both defined and equal or they are both undefined; the same can be said for $f^+(y)$ and $f^+(z)$ if $y, z \in W^s(x)$.

It is interesting to notice that $W^u(x)$ is an infinitely long line in the direction v_+ that fills densely the torus (see Problem 2.6). This implies that the collection (foliation) $\{W^u(x)\}_{x \in \mathbb{T}^2}$ of this global manifolds has a quite complex structure (see Problem 2.7). For this reason it turns out to be much more convenient to deal only with *local manifolds*.

A local manifold of size δ is simply a piece of $W^u(x)$ of size δ centered at x. In the following by $W^u(x)$ and $W^s(x)$ we will always mean local manifolds (lines) of some length. The exact length is, most of the times, irrelevant an often will not be specified (in the following it will be frequently chosen so that the lines do not wrap around the torus more than once).

Up to now we have seen that f^+ is constant along a.e. stable lines while f^- is constant along a.e. unstable line, since they are equal a.e. it seems obvious that they must be equal and constant. Yet, in the last sentence there are a lot of almost everywhere and, being measure theory a rather subtle subject, it is better to spell out the reasoning in full detail.[6]

Let us choose any point $x \in \mathbb{T}^2$ and prove that there is a neighborhood of x in which f^+ is a.e. constant. Since x is arbitrary this implies that f^+ is a.e. constant.[7] Chose a square Q_δ of size $2\delta < \frac{1}{4}$ centered at x with sides parallel to v_+ and v_- respectively. Let $\phi : [-\delta, \delta]^2 \to Q_\delta$ be defined by $\phi(\alpha, \beta) = x + \alpha v_+ + \beta v_-$, where we have chosen $\|v_\pm\| = 1$. It is then convenient to transport the problem in $[-\delta, \delta]^2$ by ϕ: doing so the Lebesgue measure is sent in the Lebesgue measure and that $f^+ \circ \phi$ is a.e. constant in the vertical direction (α constant), while $f^- \circ \phi$ is a.e. constant in the horizontal

[6] We have already seen in Examples 1.5.1–Rotations that these type of arguments must employ measure theory in a non trivial way.

[7] Please, note this apparently naïve idea to look at the problem first locally and then globally, we will see much more of it in the following.

direction. This corresponds simply to a change of variables and from now on we will confuse Q_δ and $[-\delta,\delta]^2$ since this does not create any ambiguity.

There are three full measure sets to consider:
$\widetilde{\mathcal{B}}_+ = \{\xi \in Q_\delta \mid f^+(\xi) \text{ is defined}\}$; $\widetilde{\mathcal{B}}_- = \{\xi \in Q_\delta \mid f^-(\xi) \text{ is defined}\}$ and $G = \{\xi \in \widetilde{\mathcal{B}}_+ \cap \widetilde{\mathcal{B}}_- \mid f^+(\xi) = f^-(\xi)\}$.

Let us call $W^s_\alpha := \{(a,b) \in Q_\delta \mid a = \alpha\}$ the segment in Q_δ parallel to the stable direction passing through the point $(\alpha, 0)$, and $W^u_\beta := \{(a,b) \in Q_\delta \mid b = \beta\}$ the segment in Q_δ parallel to the unstable direction passing through the point $(0,\beta)$. The previous discussion proves that there exist $\mathcal{B}_\pm \in [-\delta,\delta]$ such that $\widetilde{\mathcal{B}}_+ = \cup_{\alpha\in\mathcal{B}_+} W^s_\alpha$ and $\widetilde{\mathcal{B}}_- = \cup_{\beta\in\mathcal{B}_-} W^u_\beta$.

Since m is the product of two one dimensional Lebesgue measures[8] Fubini theorem [74] implies that $\mathcal{B}_\pm$ are measurable sets of full measure. Again by Fubini Theorem, it follows

$$4\delta^2 = m(Q_\delta) = m(\widetilde{\mathcal{B}}_+ \cap G) = \int_{\mathcal{B}_+} d\alpha \int_{-\delta}^{\delta} d\beta \chi_{W^s_\alpha \cap G}(\alpha,\beta).$$

This implies immediately that there exists a set $I \subset \mathcal{B}_+$, of full measure, such that, for each $\alpha \in I$ the set $J_\alpha = \{\beta \in \mathcal{B}_- \mid (\alpha,\beta) \in G\}$ is measurable and has full measure as well; the same holds for $E = \cup_{\alpha\in I} W^s_\alpha$.

Finally, let $z, y \in E$, $z = (a,b)$ and $y = (c,d)$. If $a = c$, then $z, y \in W^s_a$ and $f^+(z) = f^+(y)$. On the other hand, if $a \neq c$ then by choosing $\beta \in J_a \cap J_c$ it follows

$$\begin{aligned} f^+(z) &= f^+(W^s_a) = f^+(a,\beta) = f^-(a,\beta) \\ &= f^-(W^u_\beta) = f^-(c,\beta) = f^+(c,\beta) = f^+(y). \end{aligned}$$

That is, f^+ is constant on E, hence f^+ (and f^-) is a.e. constant on Q_δ. By the arbitrariness of Q_δ follows that $f^+ = f^-$ =constant a.e..

Up to now we have proved that f^+ is a.e. constant only if $f \in \mathcal{C}^{(0)}(\mathbb{T}^2)$, to prove ergodicity we need the same result for each $f \in L^1(\mathbb{T}^2)$. This can be easily obtained by an approximation argument; yet, it is probably more interesting to prove directly that all invariant sets have measure zero or one.

Let us consider a T-invariant measurable subset A. Let

$$f_n \to \chi_A \quad \text{in} \quad L^1(\mathbb{T}^2, \mu)$$

be a sequence of uniformly bounded continuous approximations to the indicator function.[9] We will use the fact that the time average is continuous with

[8] Here, to have an unambiguous notation, we should use m_n for the Lebesgue measure in $\mathbb{R}^n$, then we just said $m_2 = m_1 \times m_1$. For simplicity, I have suppressed all the subscript hoping not to confuse the reader too much.

[9] If the existence of such a sequence $\{f_n\}$ it is not obvious, consider the following: for

respect to the L^1 norm to establish that the time average of χ_A must be constant on $\mathbb{T}^2$. Indeed, if we denote by $\|\cdot\|_1$ the $L^1(\mathbb{T}^2, m)$ norm, then

$$\begin{aligned}\|f_n^+ - \chi_A^+\|_1 &= \left\| \lim_{N\to\infty} \frac{1}{N} \sum_{i=1}^N \left(f_n \circ T^i - \chi_A \circ T^i\right) \right\|_1 \\ &= \lim_{N\to\infty} \frac{1}{N} \left\| \sum_{i=1}^N \left(f_n \circ T^i - \chi_A \circ T^i\right) \right\|_1\end{aligned}$$

by the Lebesgue Dominated Convergence Theorem.

Using the invariance of the measure we obtain

$$\|f_n^+ - \chi_A^+\|_1 \leq \lim_{N\to\infty} \frac{1}{N} \sum_{i=1}^N \|f_n - \chi_A\|_1 = \|f_n - \chi_A\|_1.$$

Since the time averages $f_n^+ = m(f_n)$ a.e. in $\mathbb{T}^2$ and $\lim_{n\to\infty} m(f_n) = m(A)$, the Lebesgue dominated convergence theorem implies $\|m(A) - \chi_A^+\|_1 = 0$, that is $\chi_A^+ = m(A)$ a.e.. In addition, the invariance of A forces $\chi_A^+ = \chi_A$ so that either A or A^c has measure zero. In view of the arbitrariness of the invariant set A it follows that T must be ergodic.

2.2.2 What have we done?

The question remains of how and if such an argument can be extended to more general systems. The answer must lie in the possibility to generalize the main ingredients of the previous proof. Such ingredients are essentially two: a) the *existence* of two foliations on which f^+ (f^- respectively) are constant; b) some *regularity* property of such foliations.

In general the foliations will be provided by the stable and unstable manifolds (the existence of which is the content of the next chapter). A careful look at the previous proof should convince the reader that the needed regularity is a property of the type: consider two manifolds W_1^s, W_2^s and define a map $\phi : W_1^s \to W_2^s$ by $\phi(x) = W^u(x) \cap W_2^s$ (this is often called *holonomy map* or *Poincaré transformation*[10], we will use the first name), then ϕ is measurable and *absolutely continuous* that is : if $A \subset W_2^s$ has positive measure so has $\phi^{-1}A$. The absolutely continuity property of stable and unstable foliations will be the topic of chapter 7.

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each $\varepsilon > 0$, by the regularity of the Lebesgue measure, there exists $C_\varepsilon \subset A \subset G_\varepsilon$ (C_ε closed and G_ε open) such that $m(G_\varepsilon) - m(C_\varepsilon) \leq \varepsilon$. Then Uryshon lemma implies that there exists $f_\varepsilon \in \mathcal{C}^{(0)}(\mathbb{T}^2)$ such that $f_\varepsilon(\mathbb{T}^2) \subset [0,1]$, $f_\varepsilon|_{C_\varepsilon} = 1$ and $f_\varepsilon|_{G_\varepsilon^c} = 0$. Thus $\|f_\varepsilon - \chi_A\|_1 \leq m(G_\varepsilon \backslash C_\varepsilon) \leq \varepsilon$.

[10] Note that if one could define a flow along the unstable direction–and in our case it is possible–then the above map would indeed be a Poincaré map with respect to such a flow.

Of course, the above comments are very imprecise, their only aim is to give an idea of what is coming next. In the mean time, to start building some feeling for the foliations and their properties, see Problems 2.12, 2.13 and 2.14.

2.3 About mixing

We continue our investigations with a discussion of an other dynamical proofs in which we will see the role of hyperbolicity and some basic ideas associated to it at work. The final goal will be to obtain a dynamical proof of the following.

Proposition 2.3.1 *The Arnold cat is mixing.*

We will start by proving Topological Mixing.

Definition 2.3.2 *A smooth Dynamical System is topologically mixing if for each two open sets U and V there exists an integer $n \in \mathbb{N}$ such that*

$$T^{-m}U \cap V \neq \emptyset \quad \forall m \geq n.$$

Note that the all point in the above definition is that it holds for <u>all</u> n large enough (see Problem 2.3).

Remark that it suffices to have the above property for any class of sets that can be used as a basis for the topology. The most convenient choice is given by the so called "rectangles." Such sets are an extremely important tool in hyperbolic theory and we have already met them several times–although I will not insist on them in the present book–here they appear in the simplest possible form.

Definition 2.3.3 *By rectangle we mean a quadrilater (i.e. a region with boundaries consisting of four segments) with sides parallel to the stable or unstable directions.*

Proposition 2.3.4 *The Arnold cat is topologically mixing.*

PROOF. Let us consider two rectangles A and B. A first key observation is that, for each $m \in \mathbb{N}$, $T^m A$ and $T^m B$ are rectangles as well. The second key observation is that they have a very special shape: in the stable direction their size has contracted by a factor λ^m while in the unstable direction the size has expanded by the same factor. Hence, provided m is chosen large enough, $T^m A$ and $T^m B$ are very thin in the stable direction and very elongated in the unstable direction. This property of stretching and squeezing, that we are witnessing here, is the cornerstone of almost all arguments in hyperbolic theory. Of course, similar, but symmetrical, arguments hold for $T^{-m}A$ and $T^{-m}B$. We can then choose $m \in \mathbb{N}$ so large that the length of the unstable sides of $T^m B$ is larger than 2 and, at the same time, the same is true for the stable side of $T^{-m}A$. It is then a trivial geometric observation, best seen on the covering of $\mathbb{T}^2$, that $T^n A \cap T^{-n}B \neq \emptyset$, for each $n \geq m$, thus $T^{-2n}A \cap B \neq \emptyset$, which suffices to prove the topological mixing. □

The reader who starts to appreciate the spirit of the game may be unhappy about the previous proof. The problem is that we have used a bit too heavily the structure of the foliation (straight lines) and of $\mathbb{T}^2$ (the covering).

It is then quite natural to wonder if a more flexible and dynamical proof is available. Here it is.

Another proof of Proposition 2.3.4. Let us start by a preliminary result.

Given any rectangle A let us call A_c a rectangle of half the size and situated at its center.[11]

Lemma 2.3.5 *If $T^{-n}A_c \cap A_c \neq \emptyset$ for some $n \in \mathbb{N}$ such that $\lambda^n > 4$, then $T^{-mn}A \cap A \neq \emptyset$ for all $m \in \mathbb{N}$.*

Proof. By construction $T^{-n}A$ intersects A completely from one unstable side to the other (see figure 2.1)

This means that $T^{-2n}A \supset T^{-n}(T^{-n}A \cap A)$, which is a very thin rectangle contained in $T^{-n}A$ and that crosses it from one unstable side to the other. Accordingly $T^{-2n}A$ will intersect A completely (from one unstable side to the other). By induction the result follows. □

Note that the $n \in \mathbb{N}$ required by the above statement always exists (see Problem 2.3).

Next, let $A, B \subset \mathbb{T}^2$ be two rectangles and let $n_B \in \mathbb{N}$ such that Lemma 2.3.5 applies to B. We then consider the Dynamical Systems $(\mathbb{T}^2, T^{n_B}, m)$, this is ergodic as well (see Problem 2.2).[12] Consequently, for each integer $i \in \{1, \dots, n_B - 1\}$ there exists $k_i \in \mathbb{N}$ such that

$$T^{-k_i n_B}(T^{-i}A_c) \cap B_c \neq \emptyset,$$

and the unstable size of A times $\lambda^{-k_i n_B}$ is smaller than one quarter of the unstable size of B (see Problem 2.4). This implies immediately that

$$T^{-kn_B}(T^{-i}A) \cap B \neq \emptyset \quad \forall k \geq k_i. \tag{2.3.1}$$

[11] This may seem a silly construction but it is a rather general trick used to exploit topological mixing and we will see it again under the name of *core* of a rectangle in chapter 8.

[12] This is the crucial property always needed to obtain mixing in hyperbolic systems: ergodicity of all the powers of the map.

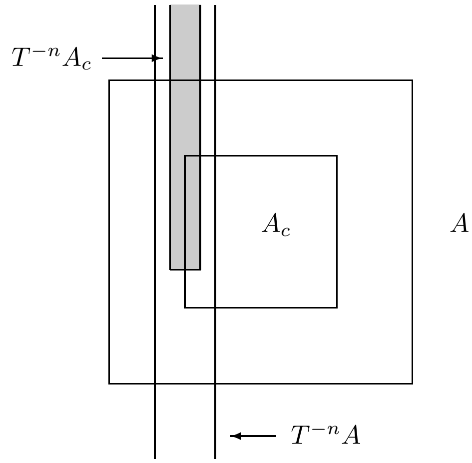

Figure 2.1: Intersection between A and $T^{-n}A$

In fact, $T^{-k_i n_B}(T^{-i}A)$ crosses B from one unstable side to the other and touches B_c, thus (2.3.1) can be proved by the same type of arguments used in Lemma 2.3.5.

Finally, set $k_m := \max\{k_i \mid i \in \{1, \dots, n_B\}\}$. For each $n > k_m n_B$ we can write $n = kn_B + i$ where $0 < i < n_B$, thus

$$T^{-n}A \cap B = T^{kn_B}(T^{-i}A) \cap B \neq \emptyset,$$

by (2.3.1). □

By the same arguments one can prove the following (see Problem 2.5).

Lemma 2.3.6 *Given any stable segment W^s of length δ, and any unstable segment W^u of length $L > \lambda\delta^{-1}$, then it holds $W^s \cap W^u \neq \emptyset$.*

To start discussing the problem of mixing we need to adopt a point of view among the many possible. We will take the one that looks at the measures (see Proposition 1.7.3 and Problem 1.29) which, by now, should be rather familiar to the reader. Calling μ_0 a measure absolutely continuous with respect to Lebesgue we would like to study the asymptotic behavior of $\mu_n := {T_*}^n \mu_0$. Thanks to Proposition 1.7.3 we need to study only the weak convergence. The first observation is that such a set of measures is compact hence we can study the set of its limit points Γ (of course with the goal of showing that it consists of only one point).[13] Such a set is simply the set of limits of

[13]Note that such accumulation points are not necessarily invariant measures, this is why we considered accumulation points of averages in section 1.4.

convergent subsequences. Since the measure μ_0 is absolutely continuous with respect to m there exists a function $h \in L^1(\mathbb{T}^2)$, $h \geq 0$, such that

$$\mu_0(f) = m(hf).$$

A lesson that we have learned from the computation in Fourier transform and from the Hopf argument is that the regularity of the functions do matter considerably and that it may be useful to consider, at first, regular functions and then obtain the wanted result by an approximation argument. Accordingly, we will restrict ourself to the case $h \in C^{(1)}(\mathbb{T}^2)$ and establish two fundamental facts.[14]

Lemma 2.3.7 *If $\bar{\mu} \in \Gamma$ then $\bar{\mu}$ is absolutely continuous with respect to Lebesgue. In addition, $\bar{h} = \frac{d\bar{\mu}}{dm} \in L^\infty(\mathbb{T}^2, m)$.*

PROOF. We notice that the sequence μ_n is uniformly absolutely continuous with respect to Lebesgue, that is $\forall f \in \mathcal{C}^{(0)}(\mathbb{T}^2)$ such that $f \geq 0$

$$\mu_n(f) = \int_{\mathbb{T}^2} h \circ T^{-n} f \leq \|h\|_\infty \|f\|_1.$$

This implies $\bar{\mu}(f) \leq \|h\|_\infty m(f)$ and

$$\bar{\mu}(A) = \sup_{\substack{C \subset A \\ C = \overline{C}}} \bar{\mu}(C) = \sup_{\substack{C \subset A \\ C = \overline{C}}} \inf_{\{f \in \mathcal{C}^{(0)} \mid f > \chi_C\}} \bar{\mu}(f) \leq \|h\|_\infty m(A), \tag{2.3.2}$$

where we have used (1.4.1) and (1.4.2). Clearly (2.3.2) implies the absolute continuity. Hence, by the Radon-Nikodym theorem [74], there exists $\bar{h} \in L^1(\mathbb{T}^2, m)$ such that $d\bar{\mu} = \bar{h} dm$.

Next, let $A = \{x \in \mathbb{T}^2 \mid \bar{h}(x) > \|h\|_\infty\}$. If $m(A) \neq 0$, then

$$\|h\|_\infty m(A) < \int_A \bar{h} dm = \bar{\mu}(A) \leq \|h\|_\infty m(A)$$

which is a contradiction, thus $\bar{h} \leq \|h\|_\infty$ a.e.. □

The next argument is very similar to what we have already seen in Examples 1.4.1–Strange Attractors. Let us call D^u the derivative along the unstable direction (if v^+ is the normal vector in the unstable direction then $D^u f := \langle \nabla f, v^+ \rangle$).

Lemma 2.3.8 *There exists $c > 0$: for each $f \in \mathcal{C}^{(1)}(\mathbb{T}^2)$*

$$|\mu_n(D^u f)| \leq \lambda^{-n} c \|f\|_\infty.$$

[14] Actually, this regularity condition on h will be needed only in Lemma 2.3.8.

Proof.

$$
\begin{aligned}
\mu_n(D^u f) &= \int_{\mathbb{T}^2} h(D^u f)\circ T^n = \int_{\mathbb{T}^2} h\langle(\nabla f)\circ T^n,\, v^+\rangle \\
&= \int_{\mathbb{T}^2} h\langle L^{-n}\nabla(f\circ T^n),\, v^+\rangle = \lambda^{-n}\sum_{i=1}^{2}\int_{\mathbb{T}^2} h\partial_{x_i}(f\circ T^n)v_i^+ \\
&= -\lambda^{-n}\int_{\mathbb{T}^2} D^u h f\circ T^n,
\end{aligned}
$$

where the last equality is obtained by integrating by parts with respect to both coordinates. Accordingly,

$$|\mu_n(D^u f)| \le \lambda^{-n}\|\nabla h\|_1\|f\|_\infty.$$

□

From the above results it follows that if $\bar\mu\in\Gamma$ then there exists $\bar h\in L^\infty(\mathbb{T}^2)$ such that, for each $f\in L^1(\mathbb{T}^2,\, m)$,

$$\bar\mu(f) = \int f\bar h dm$$

and for each $f\in\mathcal{C}^{(1)}(\mathbb{T}^2)$, $\bar\mu(D^u f)=0$. This two facts together imply that $\bar h$ is constant almost everywhere.

To see this we start by a **local** argument showing that $\bar h$ is constant along the unstable direction. We have already done a similar argument, in Examples 1.4.1–Strange Attractors, by using Fourier series, let us see here a more measure theoretical argument to convince the reader that the global structure of $\mathbb{T}^2$ has nothing to do with the result.

Let us consider an arbitrary rectangle R of size smaller than $1/4$. Consider an arbitrary $f\in\mathcal{C}^{(1)}(\mathbb{T}^2)$ with support contained in $\overset{\circ}{R}$. Then consider coordinates in R parallel to its sides (since this is achieved by rotations and rigid translations it leaves invariant the Lebesgue measure). As before, the unstable sides are horizontal. Let us call x the coordinate along the stable direction and y the one along the unstable direction. In such coordinates $R=[0,a]\times[0,b]$ (we have translates the origin at the bottom left corner of R). Given $f\in\mathcal{C}^{(1)}$, we define

$$
\begin{aligned}
\tilde f(x,\, y) &= f(x,\, y) - \frac{1}{a}\int_0^a f(\xi,\, y)d\xi, \\
F(x,\, y) &= \int_0^x \tilde f(\xi,\, y)d\xi.
\end{aligned}
$$

Then $F|_{\partial R}=0$ so F can be extended to a function on $\mathbb{T}^2$ by setting $F=0$ outside R. Note, that F is continuous and differentiable everywhere apart from the boundary ∂R where the derivative can be discontinuous. In the new coordinates D^u becomes simply the derivative with respect to x.

$$\int_{\mathbb{T}^2} \bar{h} f = \int_R \bar{h} f = \int_0^a dx \int_0^b dy \bar{h} \tilde{f} + \frac{1}{a} \int_0^b dy \int_0^a dx \bar{h}(x,\, y) \int_0^a d\xi f(\xi,\, y),$$

and, setting $\tilde{h}(y) = \frac{1}{a}\int_0^a d\xi \bar{h}(\xi,\, y)$, $\bar{f}(y) = \int_0^a d\xi f(\xi,\, y)$,

$$\int_{\mathbb{T}^2} \bar{h} f = \int_0^a dy \int_0^b dx \bar{h} \partial_x F + \int_0^b dy \tilde{h}(y) \bar{f}(y) = \int_{\mathbb{T}^2} \bar{h} D^u F + \int_0^b dy \tilde{h}(y) \bar{f}(y).$$

At this point a small obstacle appears, due to the fact that F is not $\mathcal{C}^{(1)}$. The problem is easily solved by approximating F by $\mathcal{C}^{(1)}$ functions F_ε such that $\|D^u F - D^u F_\varepsilon\|_1 \leq \varepsilon$. Then

$$\left| \int_{\mathbb{T}^2} \bar{h} D^u F \right| = \left| \int_{\mathbb{T}^2} \bar{h} D^u F - \int_{\mathbb{T}^2} \bar{h} D^u F_\varepsilon \right| \leq \|\bar{h}\|_\infty \varepsilon.$$

Hence, $\int_{\mathbb{T}^2} \bar{h} D^u F = 0$ also if the derivative is not continuous, consequently

$$\int_{\mathbb{T}^2} \bar{h} f = \int_{\mathbb{T}^2} \tilde{h} f. \tag{2.3.3}$$

By the arbitrariness of f (2.3.3) implies that $\bar{h} = \tilde{h}$ almost everywhere in $\overset{\circ}{R}$. Since R is arbitrary it follows that $\bar{h}$ is constant a.e. along the unstable direction.

A **global** argument is now needed to show that $\bar{h}$ must be constant.[15]

Proof of Proposition 2.3.1–a shortcut. Consider a line $\ell_a = \{x = a\}$. Clearly for each point $p = (a, y) \in \ell_a$ W^u_p intersects again ℓ_a at the point $(a, y + \omega_+ \mod 1)$ where $(1, \omega_+)$ is the unstable direction. Then we can consider the Dynamical Systems $(\ell_a, R_{\omega_+}, m)$, and the function $h_a = \bar{h}(a, y)$. By the previous discussion (and Fubini Theorem) it follows that, for almost every a, the function h_a is an $L^1(\ell_a, m)$ invariant function for the rotation R_{ω_+}; but we know that the irrational rotations are ergodic (see Examples 1.5.1), thus h_a =constant which implies immediately $\bar{h}$ constant. □

The above proof is simple but uses quite heavily the global properties of the foliation and of $\mathbb{T}^2$ to reduce the problem to one already studied (the irrational rotations). Clearly it is not clear how such a trick could work in more general situations. Again we would like a more flexible and dynamical argument.

[15]The fact that the argument is global, i.e. uses some properties of $\mathbb{T}^2$, reflects the fact that it is not as general has the Hopf argument which, instead, is of a completely local nature, as we will see better later (section 8.3).

PROOF OF PROPOSITION 2.3.1–DYNAMICAL. We will use a strategy already employed to prove the ergodicity of irrational rotations based on the existence of density points. Morally, this allows us to consider only rectangles. By topological mixing we can ensure that any two rectangle are crossed by the same unstable line (although it is more convenient to take preimmages of the rectangle and show that they must intersect a given unstable segment), so it is not possible that $\bar{h}$ has values different in the two rectangles. This very naïve argument can be made precise as follows.

If $\bar{h}$ it is not a.e. constant then there exists two sets A and B of positive measure such that $\bar{h}|_A > \bar{h}|_B$ a.e.. Let x_A and x_B be density points, of A and B respectively, and choose two rectangle R_A and R_B of the same size, smaller than $\frac{1}{4}$, and such that

$$\begin{aligned} m(A \cap R_A) &\geq \alpha m(R_A) \\ m(B \cap R_B) &\geq \alpha m(R_B) \end{aligned} \tag{2.3.4}$$

where $\alpha \in [0, 1)$ will be chosen later.

Let us consider $h \circ T^n$, clearly $h \circ T^n|_{T^{-n}A} > h \circ T^n|_{T^{-n}B}$ and the relations (2.3.4) hold for $T^{-n}A$, $T^{-n}R_A$ and $T^{-n}B$, $T^{-n}R_B$.

Next, let $\hat{R}_A \subset R_A$ and $\hat{R}_B \subset R_B$ be two shorter rectangles obtained by the original ones by chopping off a quarter of the length in the stable direction from each side. Let n_0 be so large that the stable length of the rectangles time λ^{n_0} is larger than one. Now chose another rectangle R, of size $\rho \leq \frac{1}{4}$, as you please. By topological mixing it follows that there exists $n > n_0$ such that $T^{-n}\hat{R}_A \cap R \neq \emptyset$ and $T^{-n}\hat{R}_B \cap R \neq \emptyset$. In addition, by the construction of $\hat{R}_A$ and $\hat{R}_B$ and the choice of n_0, it follows that $T^{-n}R_A$ and $T^{-n}R_B$ cross $\tilde{R}$ completely from one unstable side to the other, where $\tilde{R}$ is a rectangle containing R at its center and of double size. Moreover, the same quantitative argument of Lemma 2.3.6 shows that it is possible to choose n such that the stable length of $T^{-n}R_A$, $T^{-n}R_B$ is shorter than $8\lambda^2$.

Let L_A, L_B the two rectangles contained in $T^{-n}R_A \cap \tilde{R}$ and $T^{-n}R_B \cap \tilde{R}$, respectively, that cross $\tilde{R}$ from an unstable side to the other. Chose

$$\alpha = 1 - \frac{m(L_B)}{4m(R_B)} = 1 - \frac{m(L_A)}{4m(R_A)}.$$

The all point is that, on almost all the unstable lines in $\tilde{R}$, $\bar{h} \circ T^n$ is constant, so if one of this unstable lines intersects both $T^{-n}A$ and $T^{-n}B$ we have a contradiction. Thus, it must be

$$m\left(\left[\bigcup_{x \in T^{-n}A} W^u_x \cap L_B\right] \bigcap \left[\bigcup_{x \in T^{-n}B} W^u_x \cap L_B\right]\right) = 0.$$

Fubini theorem implies

$$m\left(\bigcup_{x\in T^{-n}A} W_x^u \cap L_B\right) = m\left(\bigcup_{x\in T^{-n}A} W_x^u \cap L_A\right) \geq m(T^{-n}A \cap L_A),$$

and

$$m\left(\bigcup_{x\in T^{-n}B} W_x^u \cap L_B\right) \geq m(T^{-n}B \cap L_B),$$

This yields:

$$\begin{aligned} m(L_B) &\geq m(T^{-n}A \cap L_A) + m(T^{-n}B \cap L_B) \\ &\geq m(T^{-n}A \cap T^{-n}R_A) - m(T^{-n}R_A \backslash L_A) \\ &\quad + m(T^{-n}B \cap T^{-n}R_B) - m(T^{-n}R_B \backslash L_B) \\ &\geq 2\{\alpha m(T^{-n}R_B) - m(T^{-n}R_B) + m(L_B)\} \\ &\geq \frac{3}{2} m(L_B) \end{aligned}$$

which is a contradiction. This shows that is not possible that the unstable manifolds starting at $T^{-n}A$ systematically avoid $T^{-n}B$.

Hence, $\bar{h}$ is constant, but then $\bar{h} = \int_{\mathbb{T}^2} \bar{h} = \bar{\mu}(1) = \mu_0(1)$. We have just proved that Γ consists of only one measure: the Lebesgue measure. Thus

$$\lim_{n\to\infty} \int_{\mathbb{T}^2} hf \circ T^n dm = \int_{\mathbb{T}^2} hdm \int_{\mathbb{T}^2} fdm,$$

for each $g,\, f \in C^{(1)}(\mathbb{T}^2)$. The mixing follows by the same approximation argument used in the Fourier series analyses. □

2.4 Shadowing

In this section we explore the topological complexity of the dynamics of our model systems. I have already remarked that when such a strong instability with respect to the initial condition is present it is impossible to follow exactly an orbit of the system. In fact if we compute (e.g. with a computer) the orbit of the initial point $x \in \mathbb{T}^2$, due to round off errors we do not get an orbit but rather a *pseudo-orbit*.

Definition 2.4.1 *Give an systems (X, T), X Riemannian manifold, an infinite sequence $\{x_i\}_{i\in\mathbb{Z}} \subset \mathbb{T}^2$ is called an ε-pseudo orbit if, for all $i \in \mathbb{Z}$,*

$$d(x_{i+1}, Tx_i) \leq \varepsilon.$$

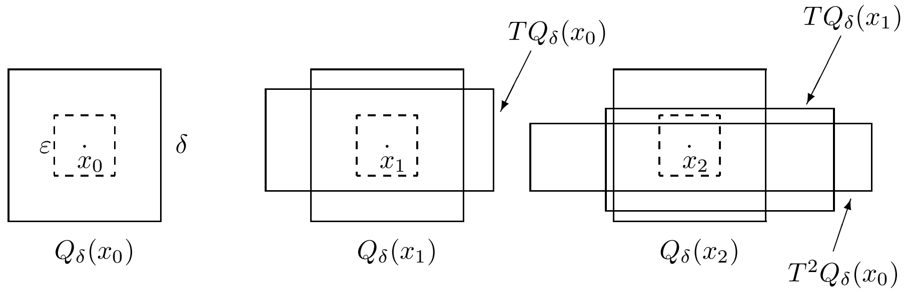

Figure 2.2: Intersection between $T^nQ_\delta(x_0)$ and $Q_\delta(x_n)$

Which means exactly that at each step an error of order ε is allowed.

The following result, beside being very useful, is a partial replay to the argument that it is not possible to follow orbits on a computer. Although the result is quite general, we state, and prove, it in our special context.

Proposition 2.4.2 *For each $\delta > 0$ there exists and $\varepsilon > 0$ such that, if $\{x_i\}$ is a ε-pseudo-orbit for the Arnold cat, then there exists $\xi \in \mathbb{T}^2$ such that*

$$d(x_i, T^i\xi) \leq \delta \quad \forall i \in \mathbb{Z}.$$

That is, there exists an orbit that δ-shadows the pseudo-orbit, moreover such an orbit is unique.

PROOF. As usual we consider rectangular (better yet: square) neighborhood of points. So, let $Q_\varepsilon(x)$ be a square of size ε centered at x with sides parallel to the stable and unstable direction, respectively.

Next, let us consider $TQ_\delta(x_0)$, since $d(Tx_0, x) \leq \varepsilon$, if $\frac{\delta}{2\lambda} + \varepsilon < \frac{\delta}{2}$ and $\frac{\lambda\delta}{2} - \varepsilon > \frac{\delta}{2}$, then $TQ_\delta(x_0)$ crosses $Q_\delta(x_1)$ completely from the stable side to the other stable side. Thus, provided we choose $\delta \geq \frac{2\lambda}{\lambda-1}\varepsilon$, we have the picture of the intersection between rectangle that we have already learned to like.

Of course the same transversal intersection takes place for each $TQ_\delta(x_i)$ and $Q_\delta(x_{i+1})$. This immediately implies that $T^nQ_\delta(x_0)$ crosses $Q_\delta(x_n)$ from one stable side to the other (see figure 2.2)

Thus $K_n = T^{-n}(T^nQ_\delta(x_0) \cap Q_\delta(x_n))$ is a sequence of nested ($K_{n+1} \subset K_n$) vertical rectangles. The unstable side of K_n is of size $\lambda^{-n}\delta$ while the stable side is of size δ.

Clearly, if $\xi \in K_n$, then

$$d(T^i\xi, x_i) < \delta \quad \forall i \in \{0, \ldots, n\}.$$

We can then consider the vertical line $K_\infty = \cap_{n\in\mathbb{N}} K_n$, by construction K_∞ consists of points whose orbit δ shadows $\{x_i\}_{i\in\mathbb{N}}$. By doing the same exact construction in the past we obtain an horizontal line $\tilde{K}_\infty$ of points that δ shadows $\{x_{-i}\}_{i\in\mathbb{N}}$. The theorem is then proven by choosing $\{\xi\} = \tilde{K}_\infty \cap K_\infty$.

the uniqueness should be obvious from the construction. In alternative the reader can prove it by contradiction. □

The above theorem is not so helpful from the measure theoretical point of view, since it could happen that the set of trajectories that shadow pseudo-orbits are of measure zero. (*say more*)

Nevertheless, it is very useful from the topological point of view (see Problem 2.15 for a dim glimpse to such possibilities).

2.5 Markov partitions

In all the above constructions the concept of rectangle has played a key rôle. In this section we present a construction that is the glorification of such a point of view.

Consider the stable and unstable manifolds of zero and prolong them until they meet (of course when they meet we meet an old friend: an homoclinic intersection) few times.

Clearly in such a way we have obtained a partition of $\mathbb{T}^2$. Such a partition consists of rectangles with sides that are either stable or unstable manifolds. We call them respectively the stable and the unstable sides of the rectangles. A partition is Markov if the preimage of each unstable side of a rectangle is contained in the unstable side of a rectangle and the image of every stable side is contained in the stable side of a rectangle. The reader can check that it is possible to use the above construction to have a Markov partition with (for example) three rectangles (see Figure 2.3 where the case $a = 1$ is drown).

Problems

2.1 Use the Diofantine properties of the stable and unstable direction to obtain better estimates of the decay of correlations. The Diofantine property refers to the following fact: if we normalize the eigenvectors in such a way that $v_\pm = (1, \omega_\pm)$, then $\omega_\pm$ are irrational numbers that are badly approximated by rationals: there exists $c \geq 0$ such that $|\omega_\pm - \frac{p}{q}| \geq \frac{c}{q^2}$ for each $p, q \in \mathbb{N}$, [1]. (Hint: ???)

2.2 Prove that the dynamical System $(\mathbb{T}^2, T^n, m)$ (where T is the Arnold cat map) is ergodic for each $n \in \mathbb{N}$. (Hint: the same proof as for $n = 1$.)

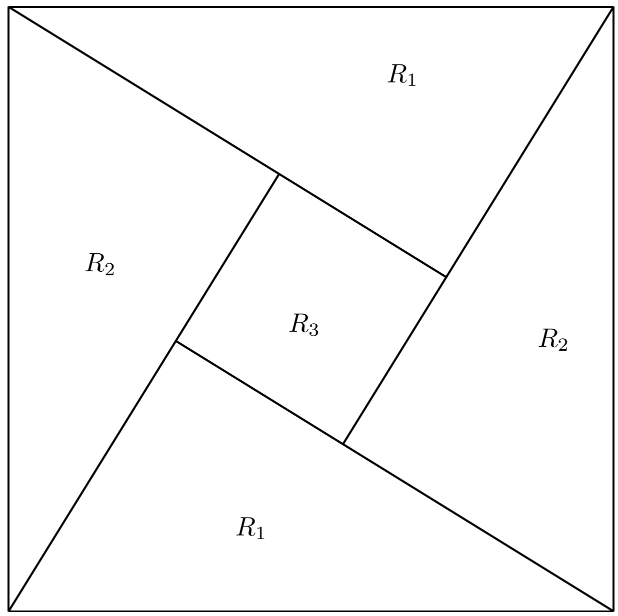

Figure 2.3: Markov partition

2.3 Let (X, T, μ) be a Dynamical Systems where X is a compact metric space, T is continuous, and μ charges the open sets (i.e. if $U \subset X$ is open, then $\mu(U) > 0$). Prove that for each $U \subset X$ open, there exist infinitely many $n \in \mathbb{N}$ such that $T^{-n}U \cap U \neq \emptyset$. (Hint: Poincaré Theorem.)

2.4 Let (X, T, μ) be an ergodic Dynamical Systems where X is a compact metric space, T is continuous, and μ charges the open sets. Prove that for each $U, V \subset X$ open, there exist infinitely many $n \in N$ such that $T^{-n}U \cap V \neq \emptyset$. (Hint:For each $k \in \mathbb{N}$, $A = \cup_{n \leq k} T^{-n}U$ is an invariant open set, if it does not intersect V, then $m(A) < 1$, thus, by ergodicity, $m(A) = 0$ which implies $U = \emptyset$.)

2.5 Prove Lemma 2.3.6. (Hint: As in the proof of Topologically mixing consider $T^{-n}W^s$, T^nW^u and chose n so large that $\lambda\delta > 2$ while the length L of W^u must satisfy $\lambda^{-n}L > 2$.)

2.6 Show that for each $x \in \mathbb{T}^2$ the global unstable manifold $W^u(x)$ is dense in $\mathbb{T}^2$. (Hint: *An algebraic proof*–Let us normalize $v_+ = (1, \omega)$, then ω is irrational. Clearly $W^u(x) = \{x + tv_+ \bmod 1\}_{t\in\mathbb{R}}$. Consider a point $y = (y_1, y_2)$ and chose $t_0 = y_1 - x_1$, then, for each $n \in \mathbb{Z}$, $x + (t_0 +$

$n)v_1^+ \bmod 1 = (y_1, R_\omega^n \xi \bmod 1)$ where $\xi = x_2 + (y_1 - x_1)\omega \bmod 1$. Now, we know that R_ω has dense orbits (see Examples 1.5.1–Rotations), thus the result.
A dynamical proof–It follows Lemma 2.3.6 plus the fact that $T^{-n}W^u$ is shorter than W^u.)

2.7 Consider the global unstable foliation $\{W^u(x)\}$ and choose an interval of length (in the horizontal direction) one from each fiber.[16] Let K be the set obtained by the union of all such segments. Prove that K it is not measurable. (Hint: Define $R : \mathbb{T}^2 \to \mathbb{T}^2$ by $R(x, y) = (x, R_\omega y)$. Then, remember Problem 1.14.)

2.8 Let $W^u(x)$, $W^s(x) \subset U \subset \mathbb{R}^2$, $U = \overset{\circ}{U}$ and $\bar{U}$ compact, smooth manifolds ($\mathcal{C}^{(1)}$ curves) such that, the $\{W^{u(s)}(x)\}_{x \in U}$ are pairwise disjoint, $\partial W^{u(s)}(x) \subset \partial U$, if $z \in W^u(x) \cap W^s(y)$, then the angle between $W^u(x)$ and $W^s(y)$ at z is larger than some $\theta > 0$. In addition, assume that, calling $v^{u(s)}(x)$ the unit tangent vector to $W^{u(s)}(x)$ at x, $v^{u(s)} \in \mathcal{C}^{(1)}$. We will call such two foliation " $\mathcal{C}^{(1)}$ uniformly transversal foliations." Show that to each such a foliation it is associated a change of variable (a diffeomorphism $\Psi : U \to U$) and that to each change of variables is associated such a foliation. (Hint: ...)

2.9 Consider two $\mathcal{C}^{(1)}$ uniformly transversal foliations (as in Problem 2.8). Prove that if $f \in L^\infty$ is constant along almost every fiber of the two foliations, then it is constant almost everywhere. (Hint: Do the argument locally and change variables so that the foliations becomes straight.)

2.10 Consider the Bernoulli measures μ_p^B defined on Σ_2^+ (the one sided sequences with two symbols) by choosing $p_0 = p$ and $p_1 = 1 - p$ (see Examples 1.1.1–Bernoulli shift). Show that, if $p \neq p'$ then μ_p^B and $\mu_{p'}^B$ are mutually singular. (Hint: All the dynamical systems $(\Sigma_2^+, \tau, \mu_p^B)$ are ergodic–See Examples ?? and ??.)

2.11 Let μ_p be the measure on $[0, 1]$ obtained from μ_p^B by the binary representation of the real numbers (see Examples), let

$$F_p(x) := \mu_p([0, x]).$$

Show that, for each $p \in (0, 1)$, $F_p : [0, 1] \to [0, 1]$ is one one, onto, continuous. In addition, show that there exists $c \in \mathbb{R}^+$ such that, for each $p, q \in [\frac{1}{4}, \frac{3}{4}]$, holds

$$|F_p(x) - F_q(x)| \leq c|p - q|.$$

[16] The Axiom of choice again.

(Hint: Note that the cylinder correspond to intervals with end points made of binary rationals. It is then immediately clear that all the measures μ_p give positive measures to the open sets. To prove the last inequality prove the representation

$$F_p(x) = \sum_{n=0}^{\infty} \sigma_n \prod_{i=0}^{n} p^{\sigma_i}(1-p)^{1-\sigma_i}$$

where σ is the binary representation of x.)

2.12 Construct $\phi : [0,1] \to [0,1]$, invertible and continuous, such that there exists $A \subset [0,1]$ with $m(A) = 0$ while $m(\phi(A)) = 1$. (Hint: Any of the above F_p will do.)

2.13 Construct a continuous foliation Ψ in $[0,1]^2$made of $\mathcal{C}^\infty$ leaves (that is Ψ is a isomomorphism of $[0,1]^2$ and $\Psi(\cdot, y) \in \mathcal{C}^\infty$). In addition, the foliation must be made of straight lines in $\{(x,y) \in [0,1]^2 \mid x \in [0, \frac{1}{4}] \cup [\frac{3}{4}, 1]\}$ but is should not be absolutely continuous in the region $\{(x,y) \in [0,1]^2 \mid x \in [\frac{1}{4}, \frac{3}{4}]\}$. (Hint: Let $\varphi \in \mathcal{C}^{(\infty)}(\mathbb{R})$, $\varphi(\mathbb{R}) = [0,1]$, $\varphi(x) = 0$ for $x < 0$ and $\varphi(x) = 1$ for $x > \frac{1}{2}$. Then, using ϕ from Problem 2.12, define

$$\Psi(x,y) = \begin{cases} (x,y) & \text{if} \quad x \in [0, \frac{1}{4}] \\ (x, [1-\varphi(x-\frac{1}{4})]y + \varphi(x-\frac{1}{4})\phi(y)) & \text{if} \quad x \in [\frac{1}{4}, \frac{3}{4}] \\ (x, \phi(y)) & \text{if} \quad x \in [\frac{3}{4}, 1]. \end{cases}$$

Clearly the leaves $\Psi(\cdot, y)$ are $\mathcal{C}^{(\infty)}$, yet the foliation it is not absolutely continuous.)

2.14 Find two $\mathcal{C}^{(0)}$ uniformly transversal foliations in $[0,1]^2$, with $\mathcal{C}^{(\infty)}$ leaves, such that the Hopf argument does not apply. (Hint: Call Ψ_p, $p \in [\frac{1}{4}, \frac{3}{4}]$ the foliation constructed in the Problem 13 starting from the function F_p defined in the Problem 11. Choose a sequence p_n converging to one quarter, e.g. $p_n = \frac{1}{4} + \frac{1}{4^n}$, then let $x_n = \frac{1}{2} - \frac{1}{2n}$. Finally define the foliation

$$\Psi(x,y) = \begin{cases} \Psi_{p_n}(x_n + (x_{n+1} - x_n)x, y) & \text{for } x \in [x_n, x_{n+1}] \\ (x, F_{\frac{1}{4}}(y)) & \text{for } x \in [\frac{1}{2}, 1] \end{cases}$$

Further define the function $g : [0,1] \to [0,1]$ to be one on a set of full measure for $\mu_{\frac{1}{4}}$ and of zero measure for μ_{p_n} and zero otherwise. The functions f^+, f^- defined by

$$f^-(x,y) = \begin{cases} 0 & \text{for } x \in [0, \frac{1}{2}) \\ 1 & \text{for } x \in [\frac{1}{2}, 1] \end{cases}$$

and

$$f^{+}(x, \Psi(x,y)) = g(x),$$

are then constant on the vertical and the Ψ foliation respectively. Moreover they clearly are equal Lebesgue almost everywhere, nevertheless they are certainly not constant.)

2.15 Show (first without using Markov Partitions and then by using Markov partitions) that the Arnold cat has at least e^{cn} periodic orbits of period n, for some $c > 0$. (Hint: If we have a rectangle R of size ε, then $T^{-n}R \cap R \neq \emptyset$ for some $n \leq c \ln \varepsilon^{-1}$. Then, if $x \in T^{-n}R \cap R$ we consider the pseudo orbit $x_k = T^i x$ where $i = k \mod n$. Then Proposition 2.4.2 implies the existence of a periodic orbit in an ε-neighborhood R_ε of R. On the other hand the boxed $T^{-k}R_\varepsilon$, $k \in \{0, \dots, n\}$ invade a part of $\mathbb{T}^2$ of measure $c\varepsilon^2 \ln \varepsilon^{-1}$. The argument is then concluded taking boxes in the remaining space and continuing until all the available space is exhausted. On the other hand, if one takes in account Markov partions, then the number of periodic orbits is given–appart from the non-invertibility of the coding–by the number of periodic simbolic sequences of period n.)

Notes

Hopf history and ref

Mention Young-Robinson example

. . . .